

Program 1

Build a Docker container from Custom Dockerfile

Project Structure

- > Dockerfile
- > requirements.txt
- > [app.py](#)

Step 1 : Create a Docker file

```
GNU nano 7.2
#Author : Uday R
#created on :

# Using Python base image
FROM python:3.9-slim

# Setting working directory
WORKDIR /app

# Copying the requirements file and install dependencies
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt

# Copying the application code
COPY . .

# Setting Flask environment variables
ENV FLASK_APP=app.py
ENV FLASK_RUN_HOST=0.0.0.0
ENV FLASK_RUN_PORT=5000

# Exposing the port
EXPOSE 5000

# Command to run flask
CMD ["flask", "run"]
```

Step 2 : Create Python file [app.py](#)

```
GNU nano 7.2
from flask import Flask

app = Flask(__name__)

@app.route("/")
def hello():
    return "Hello, Docker!"

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5000)
```

Step 3 : Create a Sample text file requirements.txt and add the following

flask

Step 4 : Execute Docker Build command

```
1rv24mc111_uday@udayraj-HP-Laptop-15s-eq2xxx:~/lab/prog1$ docker build -t prog1 .
[+] Building 7.1s (10/10) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 593B
=> [internal] load metadata for docker.io/library/python:3.9-slim
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/5] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aabb1acda1e13cf1731b1b
=> [internal] load build context
=> => transferring context: 93B
=> CACHED [2/5] WORKDIR /app
=> CACHED [3/5] COPY requirements.txt .
=> [4/5] RUN pip install --no-cache-dir -r requirements.txt
=> [5/5] COPY . .
=> exporting to image
=> => exporting layers
=> => writing image sha256:f79c5b22571323c1b39325e8c815d229e010b25e3d32478ee4238062bbbb0643
=> => naming to docker.io/library/prog1
docker:default
0.0s
0.0s
1.5s
0.0s
0.0s
0.0s
0.0s
0.0s
0.0s
5.4s
0.0s
0.1s
0.1s
0.0s
0.0s
```

Step 5 : Run the docker run command

```
1rv24mc111_uday@udayraj-HP-Laptop-15s-eq2xxx:~/lab/prog1$ docker run -d -p 5000:5000 --name cont prog1
e4ad74f55e780b0cc46808ef8ef36857e7bac165c1f56317b845264a7453f4e3
```

Step 6 : In Web Browser open <http://localhost:5000>

